TOPIC 5: MICROCOMPUTER APPLICATIONS

Official Undergraduate Lecture Slide Deck (Academic Year 2026/2027)

- Curriculum Coverage: Word Processors (5.1) | Spreadsheets (5.2) | Web & Email Protocols (5.3)
- **Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)**
- Assessment Structure: 2 Assignments (5%), 2 Tests (15%), Practical Report (10%), Practical Test (30%), UE (40%)*

**Assessment weightings are subject to adjustment by the course instructor.*

Topic 5: Lecture Outline & Learning Roadmap

Productivity Applications, Spreadsheets & Web Systems (Coverage: 5.1, 5.2, 5.3)

- Section 5.1: Word Processing Systems & Academic Document Engineering
  - Structural document architecture: Character/paragraph formatting, section breaks, margins
  - Advanced automation: Heading styles, automated Table of Contents (TOC), cross-references
  - Mail merge architecture: Data sources, merge fields, and conditional rule filters (`IF...THEN`)
- Section 5.2: Spreadsheet Applications & Quantitative Systems
  - Electronic grid architecture: Cell addressing modes (Relative, Absolute, Mixed, 3D references)
  - Function syntax & modeling: Mathematical, Logical (`IF`, `IFS`, `AND`, `OR`), Lookups (`INDEX-MATCH`, `XLOOKUP`)
  - Financial and statistical functions (`NPV`, `IRR`, `COUNTIF`, `SUMIFS`); Pivot Tables & PivotCharts
- Section 5.3: Web Browsing, Internet Protocols & Electronic Mail Architecture
  - World Wide Web: URLs, HTTP/HTTPS request-response cycles, status codes, and TLS certificates
  - Email protocols: SMTP (message submission/relay), POP3 vs IMAP4 (retrieval and sync)
  - Email security & anti-spoofing standards: SPF, DKIM, and DMARC

APPLICATION FOCUS

Master the quantitative algorithms and document automation workflows required for academic research, farm operations, and enterprise data analysis at SUA.

Topic 5: Intended Learning Outcomes (ILOs)

Core Productivity & Communications Competencies

- Engineer complex multi-section academic documents with dynamic numbering and automated tables of contents.
- Design automated mail merge pipelines for dynamic reporting and student notifications.
- Apply relative, absolute, mixed, and 3D cell referencing in complex spreadsheet calculations.
- Formulate nested logical tests, multi-condition statistical formulas, and robust lookup architectures.
- Construct dynamic business intelligence dashboards utilizing Pivot Tables, Slicers, and PivotCharts.
- Dissect Uniform Resource Locators (URLs) and trace HTTP/HTTPS client-server request-response cycles.
- Contrast POP3 and IMAP4 mail retrieval protocols and formulate SPF/DKIM anti-spoofing policies.

Word Processing: Academic Document Engineering

Beyond Basic Typing: Structural Typography and Sectioning

• Document Architecture: Hierarchical layers (Document → Sections → Paragraphs → Characters).

• Section Breaks vs Page Breaks:

- Page Break: Simply forces following text onto a new physical page; retains identical margins, orientation, and headers/footers.

- Section Break (Next Page): Divides document into independent formatting zones with distinct margins, page numbering formats, and orientations (Portrait vs Landscape).

• Thesis Compilation Rule at SUA:

- Section 1 (Preliminary Pages): Roman numerals (i, ii, iii) centered at footer.

- Section 2 (Chapter 1 onwards): Arabic numerals (1, 2, 3) aligned at top-right.

- Unlink Headers/Footers: 'Link to Previous' MUST be disabled between sections.

Automated Academic Referencing & Tables of Contents

Leveraging Semantic Heading Styles for Document Integrity

• The Semantic Heading Standard: Never manually format headings by changing font size and bolding.

• Heading Styles (Heading 1, Heading 2, Heading 3):

- Provides semantic structure recognized by document outline compilers.
- Enables one-click generation of the automated Table of Contents (TOC).

• Dynamic Updates: Modifying a chapter heading automatically updates the TOC and page numbers across a 200-page document via `F9` (Update Field).

• Table of Figures & Captions: Figures and tables tagged with official captions (`Insert Caption`), generating automated lists of figures and tables.

• Cross-Referencing: Dynamic links to numbered headings or figures that automatically track changes in figure numbering.

Mail Merge Architecture & Enterprise Dispatches

Mass Customization of Letters, Transcripts, and Advisory Notices

• Core Architecture: Combines a standardized template with a structured transactional data source.

• The Three Components of Mail Merge:

1. Main Document: Standardized text containing boilerplate prose and merge field placeholders (e.g., `«First_Name»`, `«Balance_Due»`).

2. Data Source: Structured tabular dataset (Excel spreadsheet, Access database, CSV file) where columns represent fields and rows represent records.

3. Merged Output Document: Consolidated multi-page file or direct email dispatch containing individual customized records.

• Conditional Logic Rules: Using Word `IF...THEN...ELSE` fields to dynamically insert customized advice based on numerical thresholds (e.g., 'Warning: Overdue Payment' if `Balance > 0`).

Spreadsheet Cell Addressing Paradigms

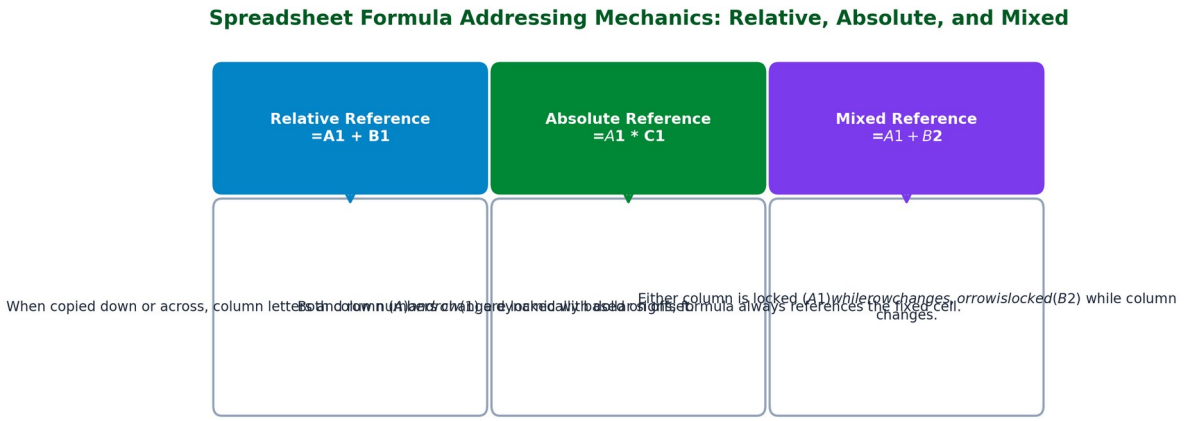


Figure: Relative, Absolute, Mixed, and 3D Spreadsheet Referencing Mechanisms.

ARCHITECTURAL ANALYSIS & PRINCIPLES

- Relative Referencing (`A1`): Offset-based; shifts row and column coordinates dynamically when copied across cells.
- Absolute Referencing (`$A$1`): Locked coordinates via `$` anchors; remains fixed regardless of copy destination.
- Mixed Referencing (`$A1` or `A$1`): Column locked and row floating, or row locked and column floating; vital for 2D multiplication tables and matrix multipliers.
- 3D Referencing (`Sheet1:Sheet4!B5`): Aggregates identical cell coordinates across multiple consecutive worksheet tabs.

Advanced Spreadsheet Functions: Logical Modeling

Automating Multi-Condition Agricultural Decisions

- Single `IF` Statement: `=IF(logical_test, value_if_true, value_if_false)`.
- Boolean Operators: `=AND(condition1, condition2)` and `=OR(condition1, condition2)`.
- Nested `IF` & Modern `IFS`:
  - Example: Cattle Weight Gain Evaluation:
  `=IFS(D2 >= 1.2, "Superior Gain", D2 >= 0.8, "Normal Gain", TRUE, "Suboptimal - Review Feed")`.
- Multi-Condition Aggregations:
  - `=COUNTIFS(Breed_Range, "Friesian", Yield_Range, ">=20")`.
  - `=SUMIFS(Milk_Liters, Farm_Block, "Mazimbu", Lactation_Stage, "Mid")`.
  - `=AVERAGEIFS(Soil_Nitrogen, Zone, "Uruguru", Crop, "Maize")`.

Lookup Architectures: VLOOKUP vs INDEX-MATCH vs XLOOKUP

The Evolution of Relational Data Retrieval in Spreadsheets

- `VLOOKUP` (`lookup_val, table_array, col_index, [range_lookup]`):
  - Severe Limitations: Can only search left-to-right (lookup column MUST be first); breaks if columns are inserted/deleted.
- `INDEX` + `MATCH` Architectural Combination:
  - `=INDEX(return_range, MATCH(lookup_val, lookup_range, 0))`.
  - Advantages: Unrestricted left-to-right or right-to-left retrieval; immune to column insertions; faster calculation on large datasets.
- `XLOOKUP` (Modern Excel Standard):
  - `=XLOOKUP(lookup_val, lookup_range, return_range, [if_not_found], [match_mode])`.
  - Combines advantages of INDEX-MATCH with concise syntax and native exact-match default.

Financial Modeling: Capital Budgeting Functions

Evaluating Agricultural Machinery Investments at SUA

- Net Present Value (`NPV`): Calculates current value of future cash inflows discounted at capital interest rate r:
  - Formula: `=NPV(rate, cash_flow_range) + initial_outlay` (Note: Initial outlay in Year 0 is added outside NPV).
- Internal Rate of Return (`IRR`):
  - Formula: `=IRR(values_range)`.
  - Calculates the exact discount rate where Net Present Value equals zero; if IRR > Cost of Capital, project is viable.
- Loan Amortization (`PMT`):
  - Formula: `=PMT(rate, nper, pv, [fv], [type])`.
  - Computes periodic loan payment for agricultural tractor financing based on constant interest rate and fixed installments.

Business Intelligence: Pivot Tables & PivotCharts

Multidimensional Summarization of Enterprise Datasets

• Pivot Table: High-speed analytical engine that groups, aggregates, cross-tabulates, and pivots large transactional records without writing complex formulas.

• Four Core Functional Quadrants:

1. Rows: Categorical variables defining horizontal row headers (e.g., Agricultural Region).
2. Columns: Categorical variables defining vertical column headers (e.g., Crop Variety).
3. Values: Numerical fields to aggregate (Sum, Average, Count, Max, Min).
4. Filters: Global dataset criteria (e.g., Harvest Year).

• Slicers & Timelines: Interactive graphical filtering buttons enabling one-click exploratory data dashboards.

• Calculated Fields: Custom mathematical formulas operating on aggregated pivot table metrics.

The World Wide Web: Architecture & URLs

The Global Distributed Hypermedia System

• Core Architecture: Client-server hypermedia infrastructure running over TCP/IP, proposed by Tim Berners-Lee (1989).

• Anatomy of a Uniform Resource Locator (URL):

`https://www.sua.ac.tz:443/academics/courses?dept=diit&level=100#syllabus`

- Protocol / Scheme: `https://` (HyperText Transfer Protocol Secure).
- Fully Qualified Domain Name (FQDN): `www.sua.ac.tz` (resolved via DNS to server IP).
- Port Number: `:443` (standard encrypted TLS port; HTTP defaults to 80).
- Path: `/academics/courses` (hierarchical resource location on web server).
- Query String: `?dept=diit&level=100` (key-value parameters passed to backend script).
- Fragment / Anchor: `#syllabus` (client-side jump target within web page).

The HTTP/HTTPS Request-Response Cycle

Stateless Communication and RESTful Operations

- Client Request: Web browser opens TCP connection (port 80/443), sends HTTP request header:
  - HTTP Methods: `GET` (retrieve resource), `POST` (submit data/payload), `PUT` (replace resource), `DELETE` (remove resource).
- Server Response: Server processes request and returns HTTP status code, response headers, and payload (HTML/JSON).
- Standard HTTP Status Code Classes:
  - 1xx (Informational): Request received, continuing process.
  - 2xx (Success): `200 OK`, `201 Created`.
  - 3xx (Redirection): `301 Moved Permanently`, `302 Found`.
  - 4xx (Client Error): `400 Bad Request`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`.
  - 5xx (Server Error): `500 Internal Server Error`, `502 Bad Gateway`, `503 Service Unavailable`.

Electronic Mail Protocol Architecture

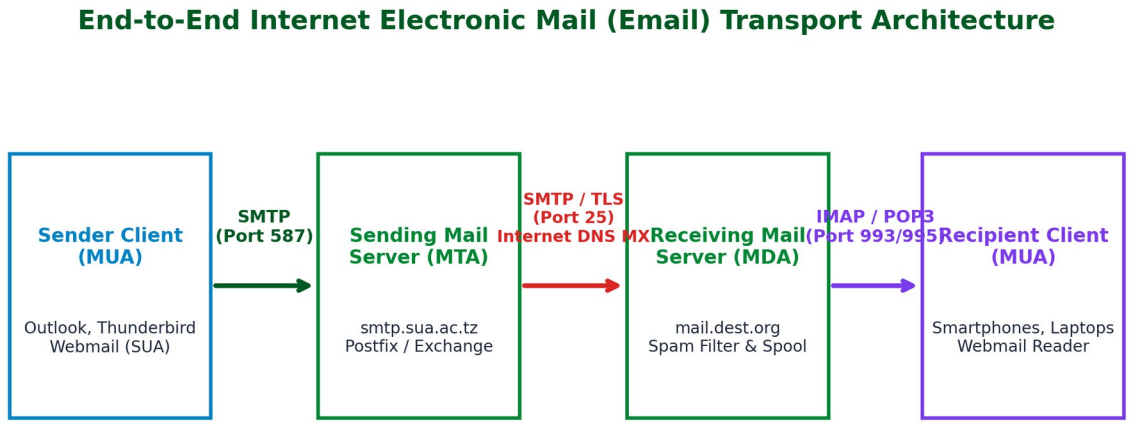


Figure: The complete email pipeline: MUA, MSA, MTA, MDA, SMTP, POP3, and IMAP4.

ARCHITECTURAL ANALYSIS & PRINCIPLES

- Sender MUA (Outlook/Thunderbird) submits email to Mail Submission Agent (MSA) via SMTP (Port 587 with STARTTLS).
- Mail Transfer Agent (MTA - Postfix/Sendmail) performs DNS MX record lookup and relays message across internet via SMTP (Port 25).
- Receiving MTA transfers message to Mail Delivery Agent (MDA - Dovecot), storing it in the user's mailbox.
- Recipient MUA retrieves or synchronizes mail via POP3 (Port 995) or IMAP4 (Port 993).

Email Retrieval: POP3 vs IMAP4

Store-and-Forward Download vs Real-Time Server Synchronization

- Post Office Protocol version 3 (POP3 - RFC 1939):
  - Ports: Port 110 (plaintext), Port 995 (POP3S encrypted via TLS).
  - Operational Model: Connects, downloads all messages to local computer disk, and deletes them from server mailbox by default.
  - Limitations: Single-device model; read status, sent items, and custom folders do NOT synchronize across other devices (smartphone, office PC).
- Internet Message Access Protocol version 4 (IMAP4 - RFC 3501):
  - Ports: Port 143 (STARTTLS), Port 993 (IMAPS encrypted via TLS).
  - Operational Model: All email messages, read/unread flags, drafts, and folder structures reside permanently on central server.
  - Advantages: Perfect multi-device synchronization; mandatory standard for modern enterprise email.

Email Anti-Spoofing Protocols: SPF, DKIM & DMARC

Combatting Domain Phishing and Fraudulent Impersonation

• The Vulnerability: Standard SMTP headers allow any sender to forge the `From:` address arbitrarily.

• 1. SPF (Sender Policy Framework - RFC 7208):

- Domain owner publishes DNS TXT record listing authorized sending mail server IP addresses.
- Example: `v=spf1 ip4:196.216.180.15 include:_spf.google.com -all`.

• 2. DKIM (DomainKeys Identified Mail - RFC 6376):

- Sending server cryptographically signs message headers with private key; recipient verifies signature using public key in sender's DNS record.

• 3. DMARC (RFC 7489):

- Coordinates SPF and DKIM alignment, instructing receiving servers what to do with failing messages (`none`, `quarantine`, `reject`) and generating forensic telemetry reports.

Academic Typography & Font Engineering

Serif vs Sans-Serif, Kerning, Tracking, and Readability

• Typeface Categories:

- Serif (e.g., Times New Roman, Georgia): Small decorative strokes at letter terminals; guides human eye across horizontal lines; traditional standard for printed body text.
- Sans-Serif (e.g., Arial, Calibri, DejaVu Sans): Clean geometric glyphs without serifs; optimal legibility on digital computer monitors and slide presentations.
- Monospaced (e.g., Courier New, Consolas): Every character occupies identical horizontal width; mandatory for code snippets, hexadecimal addresses, and tabular alignment.

• Micro-Typography:

- Leading (Line Spacing): Vertical distance between baselines (academic standard: 1.5 lines or 12 pt font with 18 pt leading).
- Kerning: Dynamic spacing adjustment between specific character pairs (e.g., 'AV', 'To').

Document Security & The PDF/A Archival Standard

Digital Preservation, Password Protection & Immutability

• Word Document Vulnerabilities: Editable `.docx` files risk accidental formula deletion, font substitution on different operating systems, and malicious macro execution.

• Portable Document Format (PDF) by Adobe (ISO 32000): Preserves exact typographical layout, vector graphics, and embedded fonts across any operating system.

• PDF/A Archival Standard (ISO 19005):

- Specifically designed for 50+ year long-term digital preservation of SUA theses and university charters.
- Strict Rules: Mandates 100% font embedding; forbids external font references, dynamic JavaScript, audio/video streams, and encryption.

• Document Protection Controls: Setting Document Open Passwords (AES-128/256) and Permission Passwords restricting printing or text copying.

Spreadsheet Data Validation & Input Sanitization

Preventing Garbage-In Garbage-Out at the Data Entry Layer

- Data Validation Engine: Enforces strict data-entry rules on worksheet cells before values are committed.
- Validation Rule Categories:
  - Whole Number / Decimal: Constrains numerical ranges (e.g., Soil pH must be between 3.5 and 9.5).
  - In-Cell Dropdown List: Restricts entries to predefined lookup tables (e.g., `Maize, Rice, Cassava, Sorghum`), eliminating typographical spelling variations.
  - Date Constraints: Ensures transaction dates fall within the active academic semester.
  - Custom Formulas: Uses boolean formulas (e.g., `=ISNUMBER(SEARCH("@sua.ac.tz", C2))` enforcing valid university email addresses).
- Error Alert Styles: Stop (blocks input completely), Warning (prompts confirmation), Information.

Conditional Formatting: Dynamic Visual Analytics

Automating Visual Anomaly Detection in Experimental Data

• Core Concept: Automatically changes cell background color, font formatting, or border based on cell values or custom evaluation formulas.

• Visualization Types:

- Data Bars: In-cell horizontal bars proportional to cell value, creating instant mini-histograms.
- Color Scales (Heatmaps): Multi-color gradients (e.g., Green-Yellow-Red) highlighting temperature or yield distribution across 100 farm plots.
- Icon Sets: Directional arrows, checkmarks, and flags indicating performance above or below benchmark thresholds.

• Formula-Driven Formatting: Highlighting an entire row if a dairy cow's status is 'Quarantine':

- Rule Formula: `=$F2 = "Quarantine"` (applies format across all columns A:G).

What-If Analysis: Goal Seek & Scenario Manager

Decision Support and Sensitivity Modeling for Agribusiness

• Goal Seek (Backwards Optimization):

- Calculates the specific input value required to achieve a desired target output value in a formula cell.
- Example: Determine the maximum tractor loan interest rate allowable to keep monthly payments below 2,500,000 TZS.

• Data Tables (Sensitivity Tables):

- 1-Variable Table: Shows how varying one input (e.g., fertilizer price) impacts multiple output metrics (profit, ROI, yield).
- 2-Variable Table: Shows how simultaneously varying two inputs (e.g., rainfall volume vs fertilizer rate) impacts total farm revenue.

• Scenario Manager: Stores multiple named sets of input assumptions (e.g., 'Drought Worst-Case', 'Average Rain', 'Abundant Harvest') and generates comparative summary reports.

Date and Time Arithmetic in Spreadsheets

Serial Day Numbers and Chronological Calculations

• The Underlying Storage Model: Excel stores dates as sequential serial integers starting from January 1, 1900 (Day 1). Today's date is an integer ~46,000.

• Time Storage: Stored as fractional decimal days (e.g., `0.5` = 12:00 PM noon; `0.75` = 6:00 PM).

• Essential Date Functions:

- `=TODAY()`: Returns current system date (volatile function).
- `=NOW()`: Returns current system date and exact time.
- `=DATEDIF(start_date, end_date, "Y")`: Computes complete elapsed years between dates (useful for animal ages or employee service length).
- `=NETWORKDAYS(start_date, end_date, [holidays])`: Calculates exact working business days, automatically excluding weekends and Tanzanian national holidays.

String Manipulation & Data Cleaning Formulas

Parsing, Cleansing, and Standardizing Raw Text Datasets

• The Raw Data Problem: Imported government or lab text files frequently contain irregular leading spaces, mixed capitalization, and concatenated identifiers.

• Text Cleansing Functions:

- `=TRIM(text)`: Strips all leading, trailing, and duplicate inner spaces.
- `=PROPER(text)` / `=UPPER(text)` / `=LOWER(text)`: Standardizes capitalization.

• Substring Extraction:

- `=LEFT(text, num_chars)` and `=RIGHT(text, num_chars)`.
- `=MID(text, start_pos, num_chars)`: Extracts middle tokens (e.g., extracting year from `SUA-2026-AGR`).
- `=LEN(text)`: Returns exact character length.

• Delimited Concatenation:

- `=TEXTJOIN(", ", TRUE, A2:D2)`: Concatenates ranges with commas, automatically skipping blank cells.

Modern Dynamic Arrays & The Spill Engine

Next-Generation Formula Execution in Microsoft Excel

- Dynamic Array Revolution: A single formula written in one cell can automatically return multiple values that 'spill' into neighboring cells.
- The `#SPILL!` Error: Occurs when the anticipated output range is blocked by existing text or values in any of the destination cells.
- Essential Dynamic Array Functions:
  - `=UNIQUE(array)`: Extracts deduplicated list of unique values (e.g., unique farmer names from 5,000 delivery rows).
  - `=FILTER(array, include_criteria, [if_empty])`: Returns rows matching specific criteria without creating pivot tables.
  - `=SORT(array, [sort_index], [sort_order])`: Sorts ranges dynamically.
  - `=SEQUENCE(rows, [cols], [start], [step])`: Generates arithmetic integer sequences instantly.

Web Browser Engine Architecture

Parsing HTML, Constructing the DOM, and JavaScript Execution

• Core Subsystems of a Modern Web Browser:

1. User Interface: Address bar, back/forward buttons, bookmarking menu.
2. Browser Engine: Coordinates actions between UI and rendering engine.
3. Rendering Engine: Parses HTML and CSS to render graphical web pages (Blink in Chrome/Edge, Gecko in Firefox, WebKit in Safari).
4. JavaScript Engine: Compiles and executes script code (V8 in Chrome, SpiderMonkey in Firefox).
5. Networking Layer: Handles HTTP/HTTPS, DNS resolution, and TLS handshakes.

• Critical Rendering Path:

- Parses HTML → Builds Document Object Model (DOM) tree.
- Parses CSS → Builds CSS Object Model (CSSOM) tree.
- Combines DOM + CSSOM → Render Tree → Layout phase → Paint phase.

Web State Management: Cookies vs LocalStorage

Maintaining Stateful Sessions Across Stateless HTTP

• HTTP is Stateless: Each HTTP request is completely independent; servers do not remember previous requests without state mechanisms.

• HTTP Cookies (RFC 6265): Small key-value text pairs (max 4 KB) stored by browser and transmitted back to server with EVERY HTTP request header.

- Session Cookies: Stored in volatile RAM; deleted when browser closes.

- Persistent Cookies: Stored on disk until expiration date (e.g., 'Remember Me' logins).

- Security Flags: `Secure` (transmitted only over HTTPS), `HttpOnly` (inaccessible to JavaScript, neutralizing XSS credential theft), `SameSite=Strict` (prevents CSRF attacks).

• HTML5 Web Storage: `localStorage` (persists indefinitely, 5–10 MB, client-side only) and `sessionStorage` (cleared on tab close).

Domain Name System (DNS) Hierarchical Resolution

Translating Human Hostnames into Machine IP Addresses

• The Phonebook of the Internet: Resolves `www.sua.ac.tz` to IPv4 `196.216.180.15`.

• The Hierarchical Tree Architecture:

1. Root DNS Servers (`.`): 13 logical root server clusters managed globally via Anycast routing.
2. Top-Level Domain (TLD) Servers: Manage gTLDs (`.com`, `.org`) and ccTLDs (`.tz`, `.ke`).
3. Authoritative Nameservers: Hold official master DNS records for the organization domain (`sua.ac.tz`).

• Essential DNS Record Types:

- `A` Record: Maps hostname to 32-bit IPv4 address.
- `AAAA` Record: Maps hostname to 128-bit IPv6 address.
- `CNAME` (Canonical Name): Alias mapping one hostname to another.
- `MX` (Mail Exchanger): Identifies mail servers responsible for domain email.
- `TXT`: Arbitrary text; houses SPF, DKIM public keys, and domain ownership tokens.

Transport Layer Security (TLS 1.3) Handshake

Establishing Encrypted, Tamper-Proof HTTPS Sessions

• Core Goals: Confidentiality (AES-GCM encryption), Integrity (HMAC authentication), Server Authenticity (X.509 digital certificate).

• TLS 1.3 Rapid Handshake Sequence (1-RTT Round Trip Time):

1. `ClientHello`: Client sends supported cipher suites and ephemeral Diffie-Hellman public key share.
2. `ServerHello`: Server selects cipher suite, sends its own key share, its X.509 certificate, and digital signature.
3. Session Key Derivation: Both parties independently calculate the shared symmetric Master Secret using Diffie-Hellman math.
4. Encrypted Transmission: All subsequent HTTP requests and responses encrypted with high-speed AES-256-GCM.

• Perfect Forward Secrecy (PFS): Even if the server's master private key is stolen years later, past recorded encrypted sessions cannot be decrypted.

Cloud Productivity Ecosystems in Academic Research

Microsoft 365, Google Workspace, and Private Clouds

• Collaborative Cloud Architecture: Real-time co-authoring using Operational Transformation (OT) and Conflict-Free Replicated Data Types (CRDT).

• Enterprise Cloud Suites:

- Microsoft 365: Word Online, Excel Online, OneDrive, SharePoint; deep enterprise Active Directory integration.
- Google Workspace: Google Docs, Sheets, Drive; streamlined browser-native collaboration.
- Self-Hosted Private Clouds (Nextcloud / ownCloud): Deployed on SUA DIIT servers to ensure sensitive national agricultural research data remains sovereign within Tanzanian borders.

• Cloud Storage Mechanics: Delta sync (transmitting only modified byte blocks of large files) and continuous versioning allowing rollback to previous revisions.

Topic 5: Key Takeaways & Synthesis Matrix

Summary of Applications & Networking for Examination Mastery

• Document Engineering: Section breaks isolate independent formatting; heading styles empower dynamic tables of contents; mail merge automates mass communications.

• Spreadsheet Referencing: Relative (`A1`) shifts dynamically; Absolute (`$A$1`) stays locked; Mixed (`$A1`/`A$1`) locks one coordinate.

• Advanced Formulas: Logical `IFS`/`COUNTIFS`/`SUMIFS` automate multi-criteria decisions; `INDEX-MATCH` and `XLOOKUP` provide flexible relational lookups.

• Web Infrastructure: URLs specify protocol, host, port, path, and query; HTTP request-response cycle handles stateless web interactions.

• Email Architecture: SMTP relays messages server-to-server; IMAP4 provides multi-device synchronized retrieval; SPF/DKIM/DMARC eliminate domain spoofing.

Review Quiz & Knowledge Check

Self-Testing Key Software & Protocol Principles

1. Why is a Section Break mandatory if you need Chapter 4 of a thesis in Landscape and Chapter 5 in Portrait?
2. When copying formula `=$C$2 * D5` from cell E5 to E6, what will the resulting formula be? (Ans: `=$C$2 * D6`).
3. Explain two major technical reasons why `INDEX` + `MATCH` is superior to `VLOOKUP` in Excel.
4. Differentiate between POP3 and IMAP4 regarding multi-device message synchronization.
5. What does the `-all` mechanism at the end of an SPF DNS TXT record instruct receiving mail servers to do?

Reading Assignments & Homework Notice

Preparation for Next Lecture & Continuous Assessment

- Required Readings:
  - Walkenbach, J. - Microsoft Excel 2019 Bible, Chapters 3, 5, 8, 11 & 14.
  - Kurose & Ross - Computer Networking: A Top-Down Approach (8th Ed.), Chapter 2 (Application Layer).
- Homework Assignment 5 Notice:
  - Available on SUA Moodle / Assignment directory: `BIT_111_Assignment_5.pdf`.
  - Total Allocation: 100 Marks (Covers Academic Document Structuring, Advanced Excel Models, and Email Protocols).
  - Submission Deadline: End of Week 10 by 17:00 EAT.
- Next Lecture Preview: Topic 6 — Data Security and Malware Defense (Physical Protection, Cryptography, Anti-Malware).